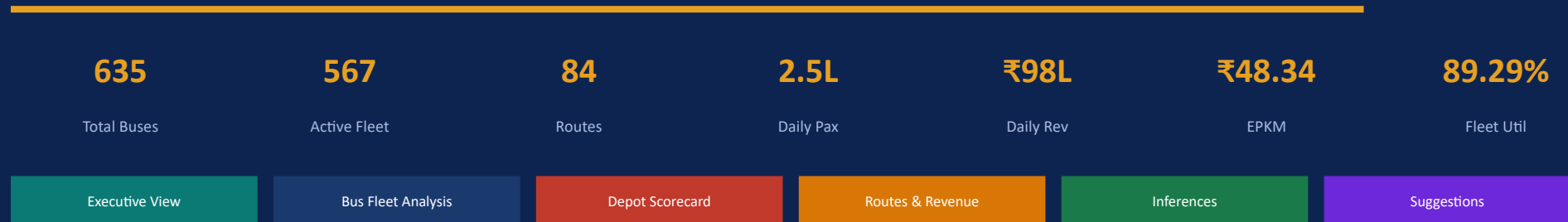


Visakhapatnam City Bus Transit

Analytics — Case Study



Power BI Analytics Dashboard • FY 2023-24

Prepared by

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635

Total City Buses

FY 2023-24

84

Active Routes

City Network

567

Effective Fleet

Operating Daily

68

Maint. Buses

Under Repair

2.5 L

Daily Passengers

+47% vs FY21-22

₹98.0 L

Daily Revenue

FY 2023-24

₹48.34

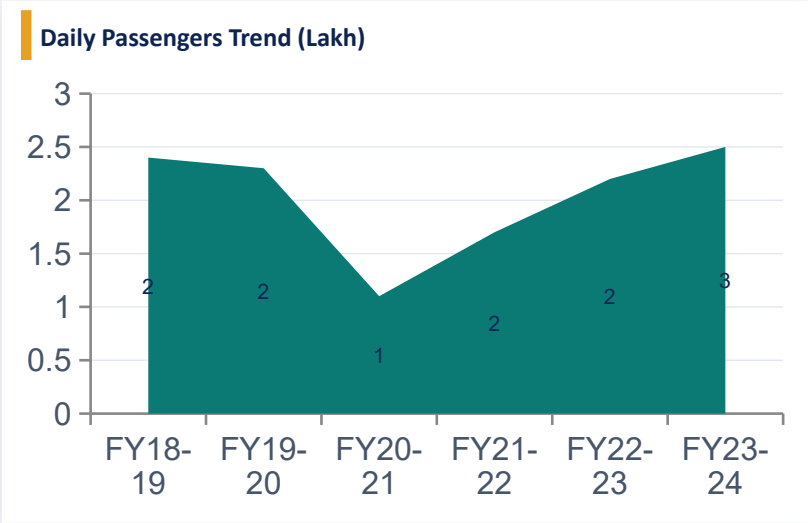
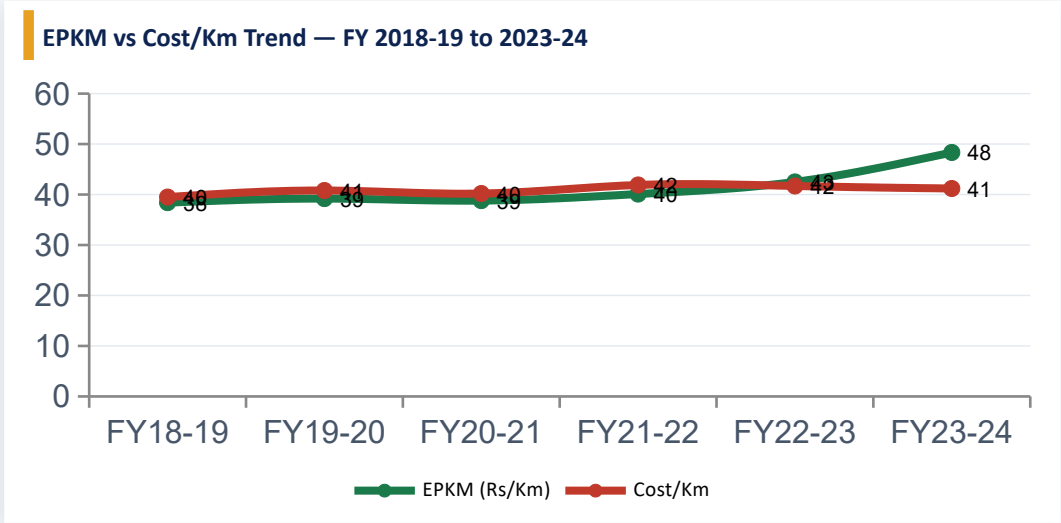
EPKM (Rs/Km)

Earnings/KM

89.29%

Fleet Utilisation

Active / Total



Bus Type vs Revenue

City Ordinary

432 buses

₹7.4M revenue

Metro Express

105 buses

₹1.8M revenue

Metro Luxury

30 buses

₹0.5M revenue

INFERENCE

APSRTC Vizag has fully recovered from COVID with 2.5L daily passengers. Crucially, EPKM (₹48.34) now exceeds Cost/Km (₹41.20) — generating a ₹7.14/km surplus. Every additional km operated is now profitable. Focus must shift from recovery to growth and efficiency.

- IMPROVEMENT OPPORTUNITY**
- Scale Metro Express & Metro Luxury: City Ordinary dominates (₹7.4M) but higher-yield premium services are underrepresented — only 135 of 567 buses.
 - Target Fleet Util >92% (currently 89.29%) by reducing dead KM through better depot scheduling.

635

Total City Buses

68

Maint. Buses

₹17,328

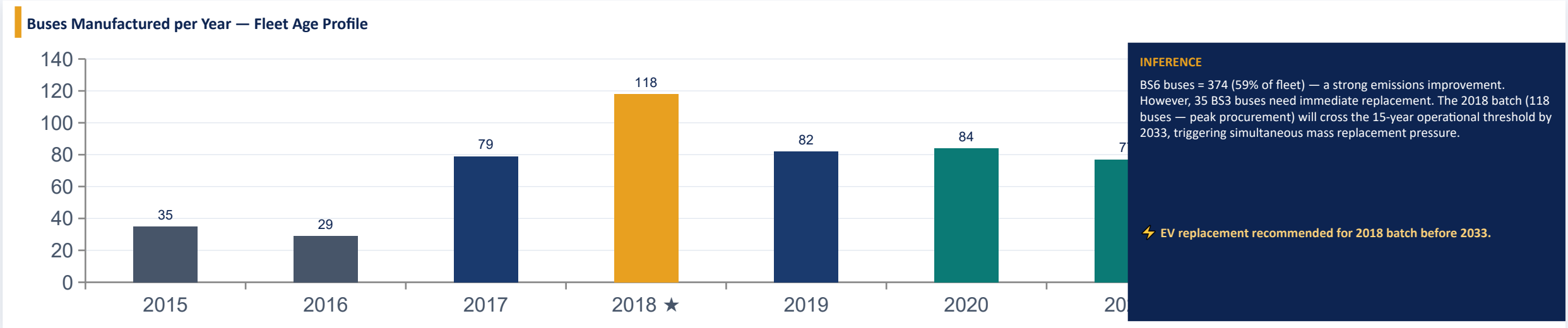
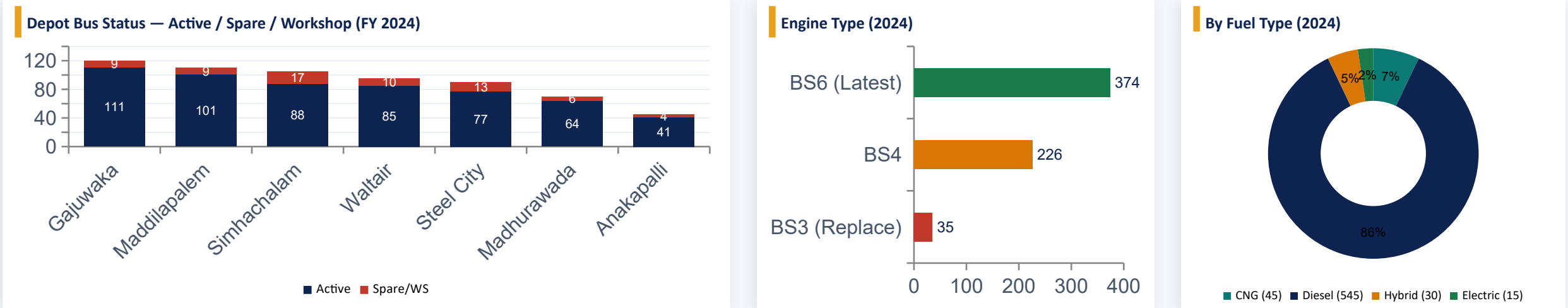
Revenue/Bus/Day

358

KM/Bus/Day

64

Fleet Age > 10 Yrs



7

Bus Depots

450

Bus Stops

₹41.20

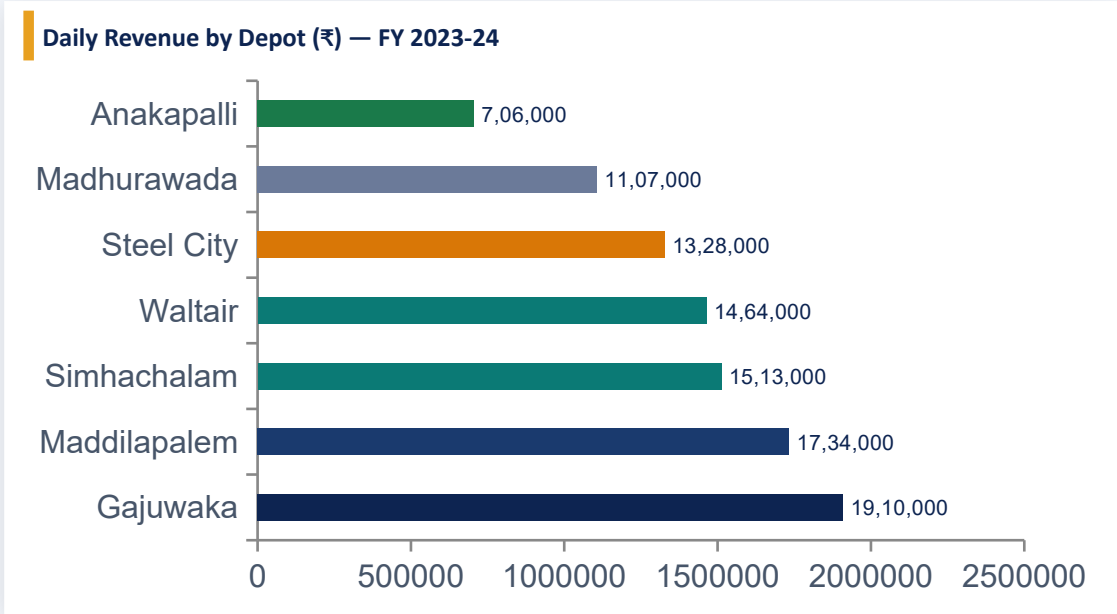
Cost / Km

16.27%

Revenue YOY Growth

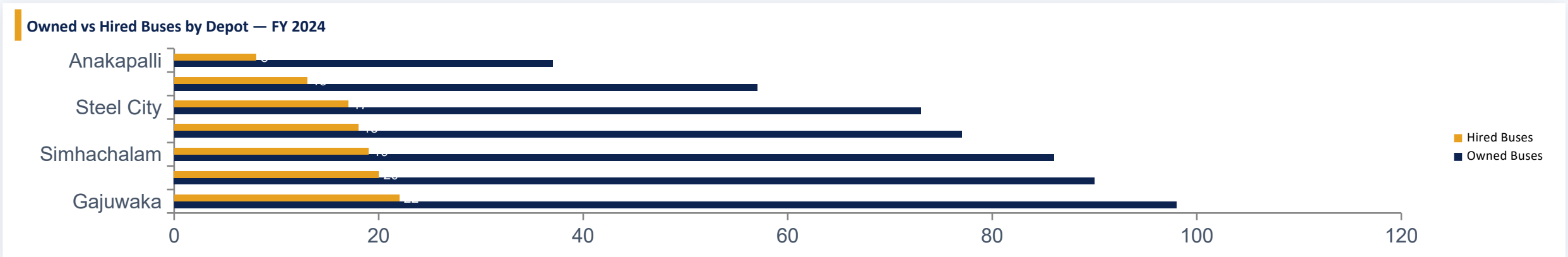
122.12%

Pax Recovery Post-Covid



Depot Performance — FY 2023-24

Depot	Buses	Pax/Day	Routes	Revenue (₹)	OTP %
Anakapalli	41	17,706	4	7,06,446	91.49 ✓
Madhurawada	64	27,480	8	11,06,783	88.40
Steel City	77	32,953	7	13,28,076	91.61 ✓
Waltair	85	36,450	12	14,64,258	80.60 ⚠
Simhachalam	88	37,804	20	15,13,237	87.20
Maddilapalem	101	43,546	24	17,33,957	82.87 ⚠
Gajuwaka	111	47,722	19	19,10,135	86.51
TOTAL	567	2,43,661	83	97,62,892	86.16



98%

GPS/AIS-140 Fitted

91.20%

ePoS Ticketing %

2,384

Daily Trips

2,054

Trips On Time

4.80 min

Avg Trip Delay

2.9%

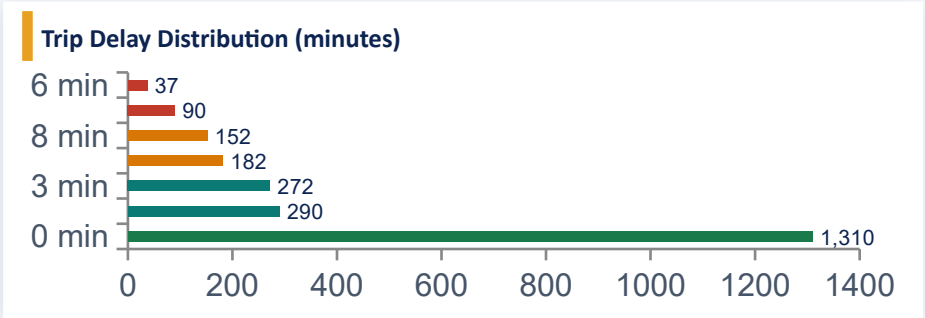
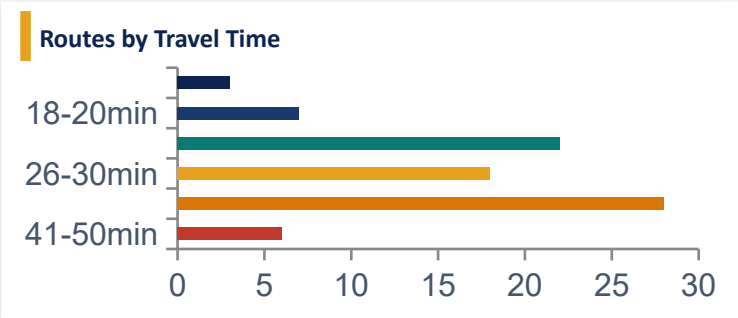
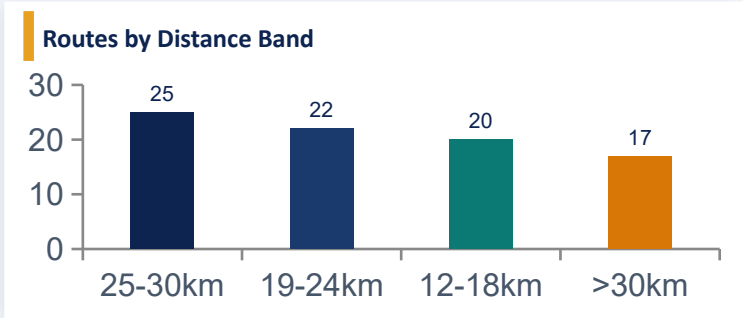
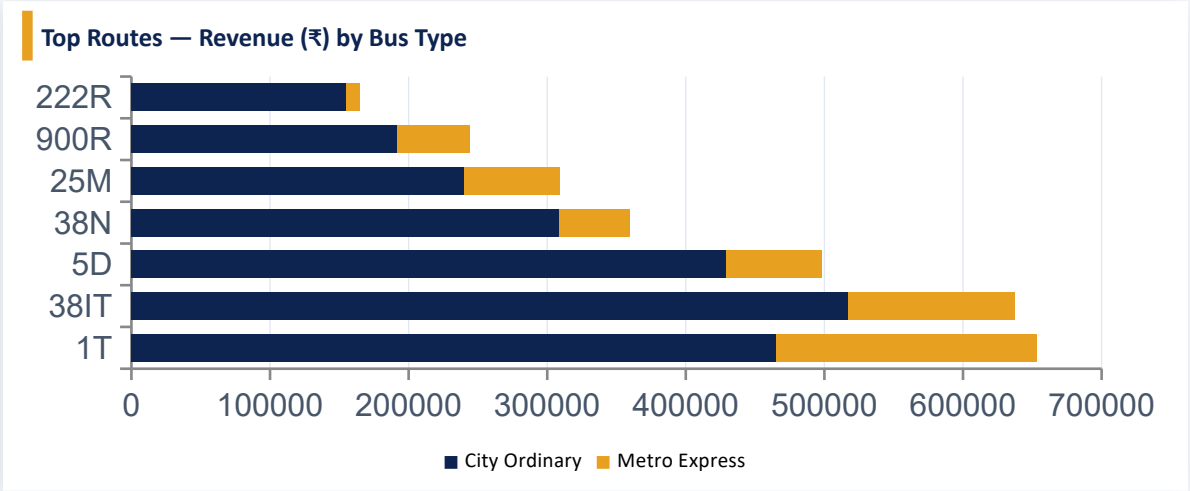
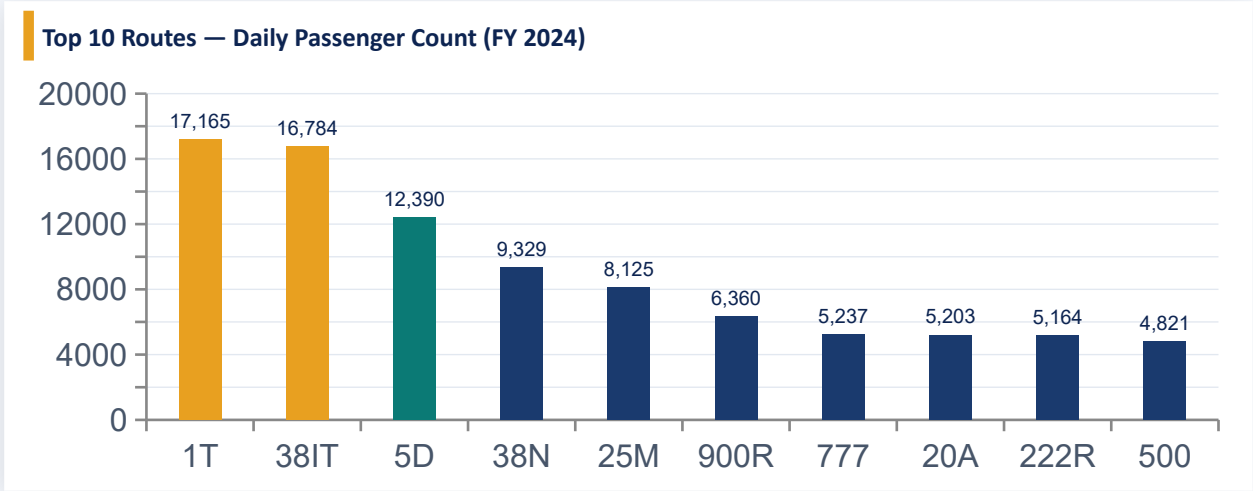
Breakdown Rate

86.16%

On-Time Perf.

25 km

Avg Route Length



Data-driven findings from all four analytical views — FY 2023-24

 01 EXECUTIVE VIEW

Full Post-Covid Recovery — Now Profitable per KM

APSRTC Vizag has recovered fully from COVID with 2.5L daily passengers in FY24. EPKM (₹48.34) now exceeds Cost/Km (₹41.20), generating a ₹7.14/km surplus. However, the original dashboard cited ₹37.6 EPKM (FY22 data) — the FY24 position is significantly stronger, confirming complete financial recovery.

Every km operated now earns ₹7.14 profit margin — operational profitability confirmed for first time post-Covid.

 03 DEPOT SCORECARD

Maddilapalem — High Load, Low On-Time Performance

Maddilapalem depot has the highest route count (24 routes), 101 buses and fleet utilisation at 89.1% — but records the lowest On-Time Performance (OTP) of 82.87% across all 7 depots. This is a structural problem: excessive route density per bus creates scheduling conflicts and cascading delays. A dedicated operational review is required.

High route count (24) per depot creates scheduling strain — consider splitting Maddilapalem operations.

 02 BUS FLEET

BS3 Fleet — Immediate Retirement Risk

BS6 buses = 374 (59% of fleet) — strong environmental compliance. However, 35 BS3 buses need immediate replacement as they violate current emission norms. The 2018 procurement batch (118 buses — highest single-year addition) will cross the 15-year service threshold by 2033. Without a phased EV transition plan starting now, APSRTC faces a simultaneous mass replacement crisis.

Plan EV induction for 118 buses (2018 batch) in phases — 2027, 2030, 2033 — to avoid fleet cliff.

 04 ROUTES & REVENUE

Routes 38IT & 1T — Maximum Revenue and Ridership Value

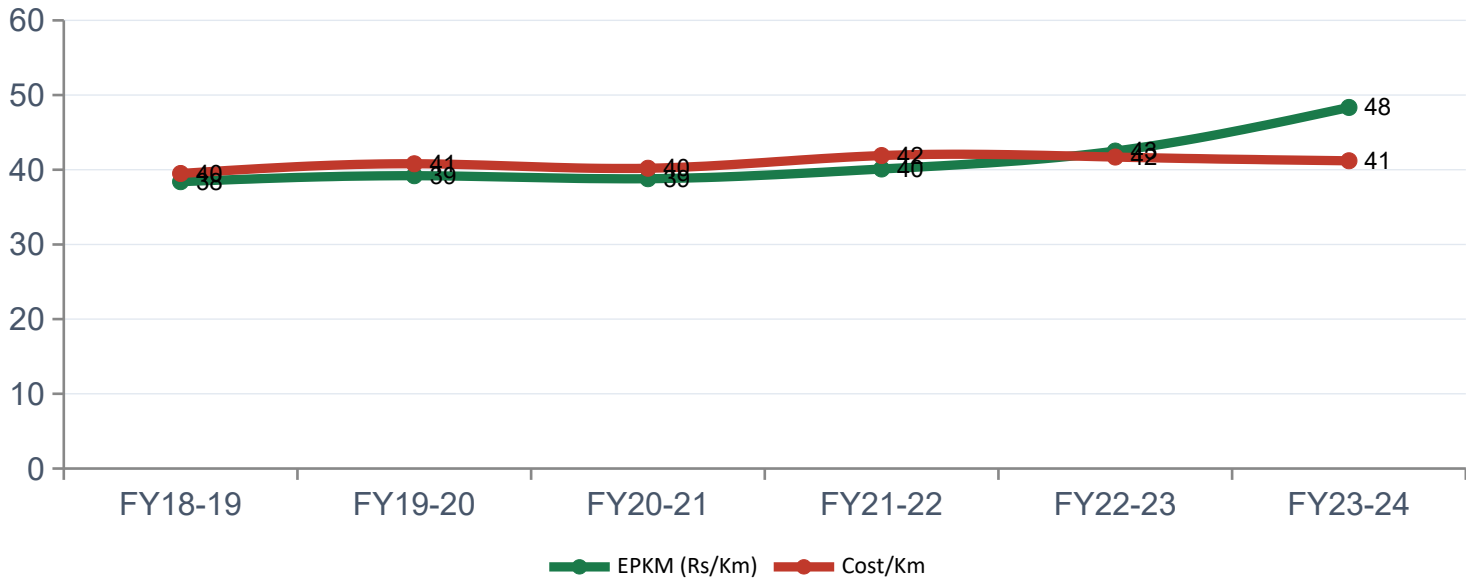
Routes 38IT and 1T lead both daily passenger count (16.8K & 17.2K) and revenue generation (₹5.17L & ₹6.53L combined). Routes 25M and 5D show comparable revenue profiles — strong candidates for frequency augmentation. Increasing bus deployment on these 4 routes delivers maximum ROI for both government revenue and public service quality.

Every additional trip on Route 1T or 38IT at ₹48.34 EPKM generates ~₹8,000–12,000 additional daily revenue.

Priority actions derived from analytical findings — structured by implementation horizon

IMMEDIATE 0 – 6 Months	SHORT TERM 6 – 18 Months	LONG TERM 18 – 36 Months
Retire 35 BS3 Buses Phase them out immediately — replace with BS6 or EV units on high-frequency routes (1T, 38IT) where passenger volumes justify the investment.	Increase Frequency on 1T, 38IT, 25M, 5D These 4 routes deliver top revenue. Add 2–3 daily trips per route — each additional trip generates ₹8K–12K extra revenue at current EPKM.	EV Fleet Transition Plan (2018 Batch) Create a phased replacement plan for 118 buses (2018 batch) expiring 2033: 40 EV buses in 2027, 40 in 2030, 38 in 2033. Apply for FAME-II / PM e-bus scheme funding.
100% ePoS Coverage Scale digital ticketing from 91.2% to 100% to eliminate residual cash revenue leakage — estimated ₹2–4L additional daily revenue recovery.	Launch Passenger-Facing App 98% of buses are GPS-fitted via AIS-140. Deploy a real-time tracking app for passengers — occupancy data from ePoS enables live seat availability display.	Aadhaar-Linked Digital Bus Pass Self-registration via Aadhaar OTP eliminates counter dependency, reduces staff costs and enables auto-renewal via UPI mandate — target 70% pass-holders migrated.
Maddilapalem OTP Fix Conduct route-by-route audit of 24 routes; reassign dedicated driver-route pairs; introduce GPS-based real-time scheduling alerts for depot supervisors.	Waltair OTP Recovery Waltair depot has 80.60% OTP — lowest in the network. Assign dedicated route controller and implement schedule cushion on peak-hour trips.	Expand Metro Express Network Only 105 of 567 buses (18.5%) are Metro Express — a higher-yield service. Target 150 Metro Express buses by FY26-27 on 12 routes with >8K daily passengers.

EPKM vs Cost/Km — Six-Year Trend (FY 2018-19 to FY 2023-24)



₹7.14/km

Profit Margin per KM
EPKM (48.34) minus Cost (41.20)

15.34%

EPKM Growth Rate
FY 2018-19 to FY 2023-24

₹98.0 L

Daily Revenue
567 buses × ₹17,328/bus/day

1.07%

Fleet CAGR
Bus count growth FY 19–24

₹41.20

Cost per KM
Operating cost benchmark

Daily Revenue Trend by Depot — FY 2023-24 (₹)



Revenue Recovery (Post-Covid)

Pax recovery rate of 122.12% — passenger volumes now exceed pre-Covid levels, confirming structural demand is intact. Revenue YOY growth at 16.27% is above inflation.

Hired Bus Cost Risk

Hired buses carry higher variable cost with no asset ownership benefit. Reducing hired fleet and replacing with owned EV buses improves long-run cost structure.

EPKM Growth Opportunity

ePoS digital ticketing at 91.2% already reduced leakage — scaling to 100% and adding Metro Express routes can push EPKM above ₹52 by FY26-27.

635

Total Buses

567

Active

84

Routes

2.5L

Daily Pax

₹98L

Revenue

₹48.34

EPKM

89.29%

Fleet Util

86.16%

OTP

Thank You

APSRTC Visakhapatnam — Moving the City Forward

Operationally Profitable

EPKM > Cost/Km for the first time post-Covid

BS3 Fleet — Act Now

35 buses need immediate replacement; plan 2018 batch EV transition

Routes 1T & 38IT

Maximum ridership & revenue — prioritise frequency augmentation

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